# Student Guide: Nix Environment Setup

Setting up reproducible course development shells with Nix and Flakes

## 1 Introduction

In this course, we use **Nix** with **Flakes** to provide a completely pre-configured, reproducible programming environment. Nix ensures that every student runs the identical compiler, runtime, libraries, and build tools as the instructor and grading infrastructure—eliminating the dreaded "it works on my machine" problem without requiring heavyweight virtual machines.

**Platform Support**

- **macOS** (Apple Silicon & Intel): Fully supported natively via Terminal.
- **Linux** (Ubuntu, Debian, Fedora, Arch, etc.): Fully supported natively.
- **Windows (via WSL2)**: Nix requires Linux. Windows students must use the **Windows Subsystem for Linux (WSL2 with Ubuntu)**. Open PowerShell as Administrator, execute `wsl -install -d Ubuntu`, restart your PC if prompted, and run all steps below inside your WSL Ubuntu terminal.

## 2 Step 1: Install Nix

Choose **Option A** (strongly recommended) or **Option B**.

### Option A (Recommended): Determinate Systems Graphical Installer

The **Determinate Systems Installer** is the easiest, most reliable way to install Nix. It provides a user-friendly graphical installer wizard, automatically enables modern features (`flakes` and `nix-command`), configures system daemons, supports macOS security settings out of the box, and includes an uninstaller.

**Recommended: Graphical Installation Steps**

1. In your web browser, navigate to: https://determinate.systems/install/
2. Click the download button for your operating system to download the graphical installer package (e.g., the `.pkg` installer on macOS).
3. Double-click the downloaded file in your `Downloads` folder to launch the graphical installation wizard.
4. Follow the on-screen prompts, click **Continue** / **Install**, and enter your system administrator password when prompted.
5. Once the installer shows the success screen, click **Close**.

### Option B (Alternative): Official Nix Installer + Flakes Configuration

If you choose the official upstream installer from the NixOS Foundation, you must explicitly enable experimental features (`nix-command` and `flakes`) in your user configuration file.

**1. Run the official multi-user daemon installer:**

```
sh <(curl -L https://nixos.org/nix/install) --daemon
```

**Required: Enable Flakes in User Configuration**

Unlike the Determinate installer, the official installer **does not** enable Flakes by default. You must configure `~/.config/nix/nix.conf`.

**2. Create your Nix user configuration directory:**

```
mkdir -p ~/.config/nix
```

**3. Append the experimental features directive:**

```
echo "experimental-features = nix-command flakes" >> ~/.config/nix/nix.conf
```

*Note: You can inspect this file with `cat ~/.config/nix/nix.conf` to ensure it contains:*
`experimental-features = nix-command flakes`

## 3 Step 2: Restart Your Terminal

After installation completes:

1. **Close your terminal completely and open a fresh terminal window.** This ensures your shell profile reloads the updated system `$PATH` pointing to Nix.
2. Verify that the Nix CLI is ready:

   ```
   nix --version
   ```

   You should see output similar to `nix (Nix) 2.x.y`.

## 4 Step 3: Navigate to Your Assignment Directory

Download or clone your assignment repository onto your machine. In your terminal, use `cd` to enter the assignment directory where `flake.nix` is located:

```
cd /path/to/your/assignment-folder
```

Confirm that the `flake.nix` configuration file exists in the directory:

```
ls -la flake.nix
```

**What is flake.nix?**

`flake.nix` defines all dependencies, compilers, libraries, test suites, and environment variables needed for the assignment.

## 5 Step 4: Launch the Development Environment (`nix develop`)

Once you have navigated into the assignment directory containing `flake.nix`, invoke:

```
nix develop
```

**What happens when you run this command?**

- **First time:** Nix evaluates `flake.nix`, downloads all pinned dependencies and tools into the isolated `/nix/store`, and configures your environment. This initial run may take a few minutes depending on your network speed.
- **Subsequent runs:** Re-entering the shell takes less than a second because all packages are already cached locally.
- **Isolated Development Shell:** You are dropped into a subshell with all project tools loaded directly into your `$PATH`. None of these tools interfere with your personal system configuration.

### Working and Exiting

Inside the shell, run your project build, test, and execution commands according to your assignment instructions. When you are finished working on the assignment, exit back to your regular shell:

```
exit
```

*You can also press `Ctrl + D` to exit.*

## 6 Summary Reference Card

| Action | Command |
|---|---|
| **Install Nix (Recommended)** | Download & run the graphical installer from https://determinate.systems/install/ |
| **Install Nix (Official)** | `sh <(curl -L https://nixos.org/nix/install) -daemon` |
| **Enable Flakes (Official)** | `mkdir -p ~/.config/nix && echo "experimental-features = nix-command flakes" >> ~/.config/nix/nix.conf` |
| **Verify Installation** | Restart terminal, then run: `nix -version` |
| **Enter Assignment Shell** | `cd /path/to/assignment` *(where flake.nix is)*<br>`nix develop` |
| **Exit Assignment Shell** | `exit` *(or press Ctrl+D)* |